

- Sector ratios (e.g., s_1/s_2 , s_3/s_4)
- Operating conditions (T, ER, SBR)
- Geometric features (d , K)
- Physical constants (gas constant R , Faraday constant F) providing dimensionality linking

3.3.3 Fibonacci Sequence Generation

Given normalized parameter N_j derived from the above sector construction:

Initialization:

$$f_1 = N_j + 2 \exp(-N_j) \quad (7)$$

$$f_2 = \exp(-N_j) \cdot f_1 \quad (8)$$

$$f_3 = f_1 + f_2 \quad (9)$$

Recursion:

$$f_n = f_{n-1} + f_{n-2} \quad \text{for } n \geq 4$$

This generates sequences encoding system state in a hierarchically structured form. Exponential damping in initialization prevents unbounded growth while preserving self-similar structure. The sequences:

- Capture both local (recent terms) and global (cumulative sum) information
- Exhibit scaling properties facilitating pattern recognition by neural networks
- Provide dimensionless representations enabling cross-case comparisons

Typically, sequences of length 10–15 suffice to capture essential dynamics without computational burden.

3.4 Neural Network Computational Laboratories

3.4.1 Architecture: Three Activation Function Laboratories

Rather than training a single large neural network, we employ three small networks—each a “computational laboratory”—with different activation functions:

1. **Linear (L):** $\phi(z) = z$ —captures linear relationships, gradient information
2. **Hyperbolic tangent (T):** $\phi(z) = \tanh(z) = (e^z - e^{-z})/(e^z + e^{-z})$ —captures smooth nonlinearities, bounded output
3. **Exponential/Sigmoid (E):** $\phi(z) = 1/(1 + e^{-z})$ —captures saturation, asymmetry

Each network receives Fibonacci sequences as input and produces a 5×5 matrix of internal weights/biases after training on a subset of gasification data. The networks are *not* used for direct prediction; instead, they serve as **statistical inference tools** extracting structure from sequences.

3.4.2 Trace Analysis and Characteristic Scoring

For each trained network and each data case, we compute:

$$\text{tr}(M) = \sum_{i=1}^5 M_{ii}$$

where M is the 5×5 internal matrix. The trace—sum of eigenvalues—is an invariant under similarity transformations, encoding aggregate system behavior without sensitivity to basis choice.

Running Fibonacci sequences from all 200+ cases through all three networks (L, T, E) in both forward (A→B) and reverse (B→A) directions generates:

$$3 \text{ networks} \times 2 \text{ directions} \times 3 \text{ configurations} = 18 \text{ traces per case}$$

These traces are consolidated into two score vectors:

$$S_{AB} \in \mathbb{R}^9 \quad (\text{forward-direction scores}) \quad (10)$$

$$S_{BA} \in \mathbb{R}^9 \quad (\text{reverse-direction scores}) \quad (11)$$

The asymmetry between forward and reverse captures **path dependence**—gasification is an irreversible thermodynamic process, and this directional asymmetry encodes hysteresis, rate limitations, and non-equilibrium effects.

3.5 Characteristic Matrix Formation

3.5.1 Outer Product Construction

Given score vectors S_{AB} and S_{BA} , we first normalize to unit length:

$$\hat{S}_{AB} = \frac{S_{AB}}{\|S_{AB}\|}, \quad \hat{S}_{BA} = \frac{S_{BA}}{\|S_{BA}\|}$$

Then form the **symmetric outer product**:

$$S_c = \frac{1}{2} \left(\hat{S}_{AB} \otimes \hat{S}_{BA}^T + \hat{S}_{BA} \otimes \hat{S}_{AB}^T \right)$$

This yields a 9×9 matrix, which is reshaped (vectorized and refolded) into a 3×3 matrix preserving essential structural information while compressing dimensionality.

In Dirac notation, this operation can be written:

$$|S_c\rangle = |S_{AB}\rangle \otimes |S_{BA}\rangle$$

emphasizing the quantum-mechanical flavor of this construction—the characteristic matrix as a tensor product state capturing correlations between forward and reverse processes.

3.5.2 Universal Structural Pattern

Remarkably, across all 200+ cases spanning diverse fuels, reactor types, and operating conditions, the characteristic matrix S_c exhibits a **consistent directional pattern**. While numerical values vary, the sign structure and dominant eigenvector orientation remain invariant—analogous to ferromagnetic ordering where local spins align to a global field.

This universality suggests S_c captures fundamental thermodynamic topology independent of specific parameter values, encoding the constraint manifold geometry onto which all gasification trajectories are confined by conservation laws.

3.6 Specific Computational Time (sct): A Novel Dynamic Property

3.6.1 Motivation: Beyond Static Equilibrium

Traditional thermodynamic properties (temperature, pressure, composition) characterize static equilibrium states. However, gasification kinetics—char reactivity, tar cracking rates, transport resistances—govern *approach* to equilibrium. How do we quantify non-equilibrium dynamics without solving time-dependent PDEs?

We introduce **specific computational time** (sct) as a proxy for “computational rigidity”—the resistance a system configuration presents to algorithmic manipulation.

3.6.2 Abstract Mechanical Operation

Consider the 3×3 characteristic matrix S_c . Define an “abstract mechanical operation”:

1. Transform matrix $\rightarrow$ vector: Vectorize S_c to 9×1 column
2. Sort ascending order: Rearrange elements from smallest to largest
3. Measure duration: Record elapsed time for this operation

Physical Analogy: This mimics material deformation under stress. “Stiff” materials (high Young’s modulus) resist deformation; similarly, certain matrix configurations may exhibit computational “stiffness”—taking longer to sort due to memory access patterns, cache misses, or algorithmic branch mispredictions.

While this may seem abstract, the key insight is that **temporal disparities in computational operations correlate with physical system complexity**. Cases requiring more intricate chemical pathways or exhibiting greater kinetic limitations manifest as matrices requiring longer sorting times.